CURRICULUM VITAE

Jeffrey Owen Hanson

Department of Biology, Carleton University, Ottawa, Canada

jeffrey.hanson@uqconnect.edu.au jeffrey-hanson.com

RESEARCH INTERESTS

My research concerns the challenges involved in conserving biodiversity. I am interested in understanding how areas can be managed to achieve conservation objectives for minimal cost. In particular, I specialize in operationalizing ecological and evolutionary processes to develop plans for protected area systems that maximize the long-term persistence of biodiversity.

EDUCATION

2013–2018 PhD, Biology, The University of Queensland, Australia (Advisors: Richard Fuller and Jonathan Rhodes, thesis: *Conserving evolutionary processes.*)

2011–2012 BSc (Hons), First Class, The University of Queensland, Australia (Advisors: Steve Salisbury, Craig Franklin, Hamish Campbell, and Ross Dwyer, thesis: *Using stable isotopes to assess the relationship between body-size, habitat use and diet in estuarine crocodiles (Crocodylus porosus)*)

2007–2010 BSc, Major in Ecology, The University of Queensland, Australia

PROFESSIONAL POSITIONS

2019–present Postdoctoral researcher at Department of Biology, Carleton University, Ottawa, Canada

2018–2019 Postdoctoral researcher at CIBIO/InBIO, Centro de Investigação em Biodiversidade e Recursos Genéticos da Universidade do Porto, Vairão, Portugal

2018 Freelance software developer contracted by Joe Bennett, University of Carleton

2012–2013 Research Assistant to Richard Fuller, The University of Queensland, Australia

2012–2013 Research Assistant to Jonathon Rhodes, The University of Queensland, Australia

2012 Casual Professional Staff, The School of Biological Sciences, The University of Queensland, Australia

PUBLICATIONS

Journal articles

2026 **Hanson JO**, McCune JL, Alamenciak T, & Bennett JR (2026) Increasing the credibility of conservation plans through citizen science. *Biological Conservation*, 313: 111552.

Castelli F, **Hanson JO**, Sabatini FM, Law E, Dutta T, & Lindner M (2026) Navigating trade-offs across ownership and ecological priorities in forest conservation planning. *Journal of Environmental Management*, 405: 129517.

Flower J, Burns ES, Dunn DC, Estep A, Everett JD, **Hanson JO**, Lester SE, & Richardson AJ (2026) oceandatr: An R package to acquire and process geospatial ocean data. *Ecology and Evolution*, 16: e74211.

Geary WL, Tulloch AIT, Doherty TS, Nimmo DG, Ritchie EG, **Hanson JO**, Maxwell MA, & Wayne AF (2026) Optimising fire and predator management for conservation. *Journal of Applied Ecology*, 63: e70256.

Jiang Z-W, Xiao H-B, **Hanson JO**, Zeng Y, Wilcove DS, Schuerch M, Kirwan ML, Yan Q-F, Chen Y, & Ma L (2026) Safeguarding global terrestrial vertebrate species from future sea-level rise. *Science*, In press: 10.1126/science.aed5772.

Khuu DT, Oldekop JA, **Hanson JO**, & Pritchard R (2026) Socially informed conservation priorities in Vietnam reveal major overlaps between biodiversity protection and human needs. *Communications Earth and Environment*, 7: 740.

2025 **Hanson JO**, Schuster R, Strimas-Mackey M, Morrell N, Edwards BPM, Arcese P, Bennett JR, & Possingham HP (2025) Systematic conservation prioritization with the prioritizr R package. *Conservation Biology*, 39: e14376.

Binley AD, **Hanson JO**, Robinson OJ, Golet GH, & Bennett JR (2025) Quantifying the value of participatory science data for conservation decision-making. *Journal of Applied Ecology*, 62: 1024–1036.

Border JA, Pearce-Higgins JW, Hewson CM, Howard C, Stephens PA, Willis, SG, Fuller RA, **Hanson JO**, Sierdsema H, Foppen RP, & Broton L (2025) Expanding protected area coverage for migratory birds could improve long-term population trends. *Nature Communications*, 16: 1–10.

Buenafe KCV, Dunn DC, Metaxas A, Schoeman DS, Everett JD, Pidd A, **Hanson JO**, Bentley LK, Kim SW, Neubert S, Scales KL, Dabalà A, Brito-Morales I, & Richardson AJ (2025) Current approaches and future opportunities for climate-smart protected areas. *Nature Reviews Biodiversity*, 1: 284–297.

Dennison CJ, Pearson AL, **Hanson JO**, Brown CD, & Buxton RT (2026) The role of vacant lots in promoting avian species diversity and occupancy in a post-industrial city. *Landscape and Urban Planning*, 265: 105518.

Giakoumi S, Richardson AJ, Doxa A, Moro S, Andrello M, **Hanson JO**, Hermoso V, Mazor T, McGowan J, Kujala H, Law E, Álvarez Romero JG, Magris RA, Gissi E, Arafeh-Dalmau N, Metaxas A, Virtanen EA, Ban NC, Runya RM, Dunn DC, Fraschetti S, Galparsoro I, Smith RJ, Bastardie F, Stelzenmüller V, Possingham HP, & Katsanevakis S (2025) Advances in systematic conservation planning to meet global biodiversity goals. *Trends in Ecology and Evolution*, 40: 395–410.

Karimi S, Schuster R, **Hanson JO**, Riva F, Liczner A, & Bennett JR (2025) Priority areas to conserve biodiversity in Canada. *FACETS*. 10: 1–10.

Jung M, Adams VM, Alagador D, Álvarez-Romero JG, Araujo MB, Arponen A, Beger M, Beher J, Carvalho SB, Giakoumi S, **Hanson JO**, Hermoso V, Jantke K, Kujala H, McGowan J, Metaxas A, O'Connor L, Salgado-Rojas J, Schuster R, Smith B, & Visconti P (2025) An interoperable and standardized protocol for reporting systematic conservation planning projects. *Conservation Science and Practice*, 7: e70097.

Morelli F, **Hanson JO**, & Benedetti Y (2025) Human pressures threaten diet-specialized mammal communities. *Proceedings of the Royal Society B: Biological Sciences*, 292: 20241735.

Neubert S, McGowan J, Metcalfe K, **Hanson JO**, Buenafe KCV, Dabalà A, Dunn DC, Everett JD, Possingham HP, Stelzenmüller V, Estep A, Ervin J, & Richardson AJ (2025) Multiple-use spatial planning for sustainable development and conservation. *Trends in Ecology and Evolution*, 40: 1126–1142.

2024 Bennett JR, Edwards BP, Bergman JN, Binley AD, Buxton RT, Hanna DE, **Hanson JO**, Hudgins EJ, Karimi S, Raymond CV, Robichaud CD, & Rytwinski T (2024) How ignoring detection probability hurts biodiversity conservation. *Frontiers in Ecology and the Environment*, 22: e2782.

Fourchault L, Dahdouh-Guebas F, Dunn DC, Everett JD, **Hanson JO**, Buenafe KCV, Neubert S, Dabalà A, Yapa KKAS, Cannicci S, & Richardson AJ (2024) Generating affordable protection of high seas biodiversity through cross-sectoral spatial planning. *One Earth*, 7: 253–264.

Hudgins EJ, **Hanson JO**, MacQuarrie CJK, Yemshanov D, Baker CM, Chadès I., Holden MH, McDonald-Madden E, & Bennett JR (2024) Spread management priorities to limit emerald ash borer (*Agrilus planipennis*) impacts on United States street trees. *Conservation Science and Practice*, 6: e13087.

Neugarten RA, Chaplin-Kramer R, Sharp RP, Schuster R, Strimas-Mackey M, Roehrdanz PR, Mulligan M, van Soesbergen A, Hole D, Kennedy CM, Oakleaf

JR, Johnson JA, Kiesecker J, Polasky S, **Hanson JO**, & Rodewald AD (2024) Mapping the planet's critical areas for biodiversity and nature's contributions to people. *Nature Communications*, 15: 261.

2023 **Hanson JO**, McCune JL, Chadès I, Proctor CA, Hudgins EJ, & Bennett JR (2023) Optimizing ecological surveys for conservation. *Journal of Applied Ecology*, 60: 41–51.

Buenafe KCV, Dunn DC, Everett JD, Brito-Morales I, Schoeman DS, **Hanson JO**, Dabalà A, Neubert S, Cannicci S, Kaschner K, & Richardson AJ (2023) A metric-based framework for climate-smart conservation planning. *Ecological Applications*, 33: e2852.

Chowdhury S, Fuller RA, Rokonuzzaman M, Alam S, Das P, Siddika A, Ahmed S, Labi MM, Chowdhury SU, Mukul SA, Bohm M, & **Hanson JO** (2023) Insights from citizen science reveal priority areas for conserving biodiversity in Bangladesh. *One Earth*, 6: 1315–1132.

Chowdhury S, Zalucki MP, **Hanson JO**, Tiatragul S, Green D, Watson JEM, & Fuller RA (2023) Three quarters of insects are insufficiently covered by protected areas. *One Earth*, 6: 1–8.

Dabalà A, Dahdouh-Guebas F, Dunn DC, Everett JD, Lovelock CE, **Hanson JO**, Buenafe KVC, Neubert S, & Richardson AJ (2023) Priority areas to protect mangroves and maximise ecosystem services. *Nature Communications*, 14: 5863.

Justeau-Allaire D, **Hanson JO**, Lannuzel G, Vismara P, Lorca X, & Birnbaum, P (2023) restoptr: an R package for ecological restoration planning. *Restoration Ecology*, 31: e13910.

Schuster R, Buxton R, **Hanson JO**, Binley AD, Pittman J, Tulloch V, La Sorte FA, Roehrdanz PR, Verburg PH, Rodewald AD, Wilson S, Possingham HP, & Bennett JR (2022) Protected area planning to conserve biodiversity in an uncertain future. *Conservation Biology*, 37: e14048.

Shen X, Liu M, **Hanson JO**, Wang J, Locke H, Watson JEM, Ellis EC, Li S, & Ma K (2023) Countries' differentiated responsibilities to fulfill area-based conservation targets of the Kunming-Montreal Global Biodiversity Framework. *One Earth*, 6: 548–559.

2022 **Hanson JO**, Vincent J, Schuster R, Fahrig L, Brennan A, Martin AE, Hughes JS, Pither R, & Bennett JR (2022) A comparison of approaches for including connectivity in systematic conservation planning. *Journal of Applied Ecology*, 59: 2507–2519.

Hanson JO (2022) wdpar: Interface to the World Database on Protected Areas. *Journal of Open Source Software*, 7: 4594.

Carter ZT, **Hanson JO**, Perry GLW, & Russell JC (2022) Incorporating management action suitability in conservation plans. *Journal of Applied Ecology*, 59: 2581–2592.

Marques AJD, **Hanson JO**, Camacho-Sanchez M, Martínez-Solano Í, Moritz C, Tarroso P, Velo-Antón G, Veríssimo A, & Carvalho SB (2022) Range-wide genomic scans and tests for selection identify non-neutral spatial patterns of genetic variation in a non-model amphibian species (*Pelobates cultripes*). *Conservation Genetics*, 23: 387–400.

Nielsen ES, **Hanson JO**, Carvalho SB , Beger M, Henriques R, Kershaw F, & von der Heyden S (2022) Molecular ecology meets systematic conservation planning. *Trends in Ecology & Evolution*, 38: 143–155.

2021 **Hanson JO**, Veríssimo A, Velo-Antón G, Marques A, Camacho-Sanchez M, Martínez-Solano Í, Gonçalves H, Sequeira F, Possingham HP, & Carvalho SB (2021) Evaluating surrogates of genetic diversity for conservation planning. *Conservation Biology*, 35: 634–642.

Jung M, Arnell A, de Lamo X, García-Rangel S, Lewis M, Mark J, Merow C, Miles L, Ondo I, Pironon S, Ravilious C, Rivers M, Schepashenko D, Tallowin O, van Soesbergen A, Govaerts R, Boyle BL, Enquist BJ, Feng X, Gallagher R, Maitner B, Meiri S, Mulligan M, Ofer G, Roll U, **Hanson JO**, Jetz W, Di Marco M, McGowan J, Rinnan DS, Sachs JD, Lesiv M, Adams VM, Andrew SC, Burger JR, Hannah L, Marquet PA, McCarthy JK, Morueta-Holme N, Newman EA, Park DS, Roehrdanz PR, Svenning J-C, Violle C, Wieringa JJ, Wynne G, Fritz S, Strassburg BBN, Obersteiner M, Kapos V, Burgess N, Schmidt-Traub G, & Visconti P (2021) Areas of global importance for conserving terrestrial biodiversity, carbon and water. *Nature Ecology & Evolution*, 5: 1499–1509.

Morelli F, Benedetti Y, **Hanson JO**, & Fuller RA (2021) Global distribution and conservation of avian diet specialization. *Conservation Letters*, 14: e12795.

Naia M, Tarroso P, Sow AS, Liz AV, Gonçalves DV, Martínez-Freiría F, Santarém F, Yusefi GH, Velo-Antón G, Avella I, **Hanson JO**, Khalatbari L, Ferreira da Silva MJ, Camacho-Sanchez M, Boratyński Z, Carvalho SB, & Brito JC (2021) Potential negative effects of the Green Wall on Sahel's biodiversity. *Conservation Biology.* 35: 1966–1968.

2020 **Hanson, JO**, Rhodes JR, Butchart SHM, Buchanan GM, Rondinini C, Ficetola GF, & Fuller RA (2020) Global conservation of species' niches. *Nature*, 580: 232–234.

Hanson JO, Marques A, Veríssimo A, Camacho-Sanchez M, Velo-Antón G, Martínez-Solano Í, & Carvalho SB (2020) Conservation planning for adaptive and neutral evolutionary processes. *Journal of Applied Ecology*, 57: 2159–2169.

Camacho-Sanchez M, Velo-Antón G, **Hanson JO**, Veríssimo A, Martínez-Solano Í, Marques A, Moritz, C & Carvalho SB (2020) Comparative assessment of range-wide patterns of genetic diversity and structure with SNPs and microsatellites: A case study with Iberian amphibians. *Ecology & Evolution*, 10: 10353–10363.

Schuster R, **Hanson JO**, Strimas-Mackey M, & Bennett JR (2020) Exact integer linear programming solvers outperform simulated annealing for solving conservation planning problems. *PeerJ*, 8: e9258.

2019 **Hanson JO**, Fuller RA, & Rhodes JR (2019) Conventional methods for enhancing connectivity in conservation planning do not always maintain gene flow. *Journal of Applied Ecology*, 56: 913–922.

Hanson JO, Schuster R, Strimas-Mackey M, & Bennett JR (2019) Optimality in prioritizing conservation projects. *Methods in Ecology & Evolution*, 10: 1655–1663.

Ambrose L, **Hanson JO**, Riginos C, Xu W, Fordyce S, Cooper RD, & Beebe NW (2019) Population genetics of *Anopheles koliensis* through Papua New Guinea: New cryptic species and landscape topography effects on genetic connectivity. *Ecology & Evolution*, 9: 13375–13388.

2018 **Hanson JO**, Rhodes JR, Possingham HP, & Fuller RA (2018) raptr: Representative and Adequate Prioritization Toolkit in R. *Methods in Ecology & Evolution*, 9: 320–330.

2017 **Hanson JO**, Rhodes JR, Riginos C, & Fuller RA (2017) Environmental and geographic variables are effective surrogates for genetic variation in conservation planning. *Proceedings of the National Academy of Sciences of the United States of America*, 114: 12755–12760.

Mather AT, **Hanson JO**, Pope LC, & Riginos C (2017) Comparative phylogeography of two co-distributed but ecologically distinct rainbowfishes of far-northern Australia. *Journal of Biogeography*, 45: 127–141.

2016 Dudaniec RY, Worthington Wilmer J, **Hanson JO**, Warren M, Bell S, & Rhodes JR (2016) Dealing with uncertainty in landscape genetic resistance models: a case of three co-occurring marsupials. *Molecular Ecology*, 25: 470–486.

Dhanjal-Adams KL, **Hanson JO**, Murray NJ, Phinn SR, Wingate VR, Mustin K, Lee JR, Allan JR, Cappadonna JL, Studds CE, Clemens RS, Roelfsema CM, & Fuller RA (2016) Distribution and protection of intertidal habitats in Australia. *Emu*, 116: 208–214.

2015 **Hanson JO**, Salisbury SW, Campbell HA, Dwyer RG, Jardine TD & Franklin CE (2015) Feeding across the food web: the interaction between diet, movement and body size in estuarine crocodiles (*Crocodylus porosus*). *Austral Ecology*, 40: 275–286.

Runge CA, Watson JEM, Butchart HM, **Hanson JO**, Possingham HP, & Fuller RA (2015) Protected areas and global conservation of migratory birds. *Science*, 350: 1255–1258.

Auerbach NA, Wilson KA, Tulloch AI, Rhodes JR, **Hanson JO**, & Possingham HP (2015) Effects of threat management interactions on conservation priorities. *Conservation Biology*, 29: 1626–1635.

Bunton JD, Ernst AT, **Hanson JO**, Beyer HL, Hammill E, Runge CA, Venter O, Possingham HP, & Rhodes JR (2015) Integrated planning of linear infrastructure and conservation offsets. In Weber, T, McPhee, MJ, & Andersson RS (eds) *MODSIM 2015, 21st International Congress on Modelling and Simulation.* Modelling and Simulation Society of Australia and New Zealand, December 2015, pp. 1427–1433.

Rabeb D, Othman DS, Essilfie AT, Hansbro PM, **Hanson JO**, McEwan AG, & Kappler U (2015) Maturation of molybdoenzymes and its influence on the pathogenesis of non-typeable *Haemophilus* influenzae. *Frontiers in Microbiology*, 6: 01219.

Non-peer reviewed articles

2020 **Hanson JO** (2020) Conserving species' niches. *Nature Ecology & Evolution Community*, https://go.nature.com/2wCckAN.

2015 Beher J & **Hanson JO** (2015) Welcome to the Mapotron. *Decision Point*, **86**: 10–11.

PRESENTATIONS

Conference presentations

2025 **Hanson JO**, Delsen DM, Binley A, Allan J, Jung M, Visconti P, Hermoso V, Schuster R, Chapman M, & Bennett JR (2025) *Optimally managing threats to biodiversity across large scales.* Oral presentation to 32nd International Congress for Conservation Biology, Brisbane, Australia.

2022 **Hanson JO**, Delsen DM, Binley A, Allan J, Jung M, Visconti P, Hermoso V, Schuster R, Chapman M, & Bennett JR (2022) *Optimally managing threats to biodiversity across large scales.* Oral presentation to New Zealand Ecological Society, Dunedin, New Zealand.

Hanson JO, McCune JL, Chadès I, Proctor CA, Hudgins EJ, & Bennett JR (2022) *Optimizing ecological surveys for conservation.* Oral presentation to 2022 Annual Meeting of Ecological Society of America, Montreal, Canada.

2019 **Hanson JO**, Camacho-Sanchez M, Marques A, Martínez-Solano Í, Velo-Antón G, Veríssimo A, & Carvalho SB (2019) *Conserving evolutionary processes for three am-*

phibian species in the Iberian Peninsula. Oral presentation to European Ecological Federation 15th Congress, Lisbon, Portugal.

2018 **Hanson JO**, Schuster R, Morrell N, Strimas-Mackey M, Watts ME, Arcese P, Bennett JR, & Possingham HP (2018) *prioritizr: Systematic conservation prioritization in R.* Oral presentation to UseR! 2018, Brisbane, Australia.

2016 **Hanson JO**, Rhodes JR, Possingham HP, & Fuller RA (2016) *RAPR: Representative and Adequate Prioritizations in R.* Oral presentation to Society for Conservation Biology 4th Oceania Congress, Brisbane, Australia.

2014 **Hanson JO**, Rhodes JR, & Fuller RA (2014) *Conservation planning for intraspecific biodiversity using surrogates.* Oral presentation to the Meeting of the Minds mini-conference at The University of Queensland, Brisbane, Australia.

PEER REVIEW ACTIVITIES

I have reviewed submissions to following journals: *Austral Ecology*, *Conservation Letters*, *Diversity and Distributions*, *Ecological Applications*, *Global Change Biology*, *Journal of Applied Ecology*, *Methods in Ecology & Evolution*, *PLoS ONE*, *Nature Ecology & Evolution* and *Proceedings of the Royal Society of London B: Biological Sciences*.

EDUCATIONAL ACTIVITIES

Classroom Instruction, The University of Queensland, Australia

2013–2015 Professional tutor to the "Field Ecology" course, coordinated by Myron Zalucki

Workshop Instruction

2025 *Conservation planning in R: Spatial Data Wrangling and Spatial Prioritizations* coordinated by Sandra Neubert, Kristine Buenafe, and Alvise Dabalà at the 32nd International Congress for Conservation Biology, Brisbane, Australia.

2019 *Spatial Conservation Prioritization: Concepts, Methods and Applications* coordinated by Silvia Carvalho, Virgilio Hermoso, and Jeffrey Hanson at CIBIO/InBIO, Centro de Investigação em Biodiversidade e Recursos Genéticos da Universidade do Porto, Vairão, Portugal.

2017 *Use of Machine Learning in Conservation, Moving beyond just Maxent and SDMs* coordinated by Falk Huettmann at the 28th International Congress for Conservation Biology (ICCB), Cartagena, Colombia.

2015 *Geospatial Analysis in R* coordinated by Hawthorne Beyer, Rebbecca Runting and Jutta Beher at the Student Conference of Conservation Science, Australia.

Smoothing the Marxan Flow with R coordinated by Matthew Watts at the Student Conference of Conservation Science, Australia.

2013 *Introduction to Geospatial Analysis* coordinated by Hawthorne Beyer at The University of Queensland, Australia.

Introduction to Spatial Data Analysis in R workshop coordinated by Hawthorne Beyer at The University of Queensland, Australia.

2011 *Introducing R* coordinated by Simon Blomberg at The University of Queensland, Australia.

Seminars

2021 *Making better conservation decisions* presented online to members of the Center for Biodiversity and Global Change at Yale University, The United States of America.

Making better conservation decisions presented to members of the School of Agriculture and Environment at Massey University, New Zealand.

2017 *Systematic conservation prioritization in R* presented to members of the Center for Biodiversity and Conservation Science at The University of Queensland, Australia.

2016 *Biodiversity processes in reserve-selection* presented to members of the Center for Biodiversity and Conservation Science at The University of Queensland, Australia.

SOFTWARE DEVELOPMENT

prioritizr An R package for building and solving conservation planning problems (https://CRAN.R-project.org/package=prioritizr).

Where to Work An interactive decision support tool for conservation planning (https://ncc.carleton.ca).

oppr An R package for prioritizing funding for threatened species recovery projects and pest management project (https://CRAN.R-project.org/package=oppr).

wdpar An R package for downloading and cleaning data from the World Database on Protected Areas (https://CRAN.R-project.org/package=wdpar).

surveyvoi An R package for optimizing survey design for conservation decision making (https://CRAN.R-project.org/package=surveyvoi)

raptr An R package for generating spatial prioritizations that secure intra-specific variation (https://CRAN.R-project.org/package=raptr).

TECHNICAL SKILLS

Programming languages R, C++, JavaScript.

Data analysis spatial prioritization, species distribution models, tidyverse, general linear models, generalized linear mixed effects models, gradient boosted regression trees, cluster analysis.

Development frameworks bookdown, shiny, golem, htmlwidgets, Rcpp, make.

Geographic information systems sf, terra, raster, GDAL.

Continuous integration/deployment Docker, GitHub Actions, shinyproxy.

Operating systems Ubuntu, Microsoft Windows.

Miscellaneous rmarkdown, LaTeX.

MEETINGS AND NETWORKING EVENTS

2018 rOpenSci Unconference, Melbourne, Australia.

2017 R Unconference hosted by the Brisbane Users of R Group, Brisbane, Australia.

2016 rOpenSci Unconference, Brisbane, Australia.

SCHOLARSHIPS AND AWARDS

- Compute resource allocation by the National eResearch Collaboration Tools and Resources (NeCTAR) project (2017–2019)
- Postgraduate Travel Award Scholarship, The School of Biological Sciences, The University of Queensland, Australia (2016)
- Australian Postgraduate Award (APA) Scholarship (2013)